CREDIT CARD FRAUD DETECTION PROJECT

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OBJECTIVES & ACHIEVEMENTS

- Objective: Deploy a proactive machine learning solution to mitigate fraud losses while maintaining high customer satisfaction.

- Key Achievement: 81.5% reduction in fraud-related costs

UNDERSTANDING THE PROBLEM & DATA

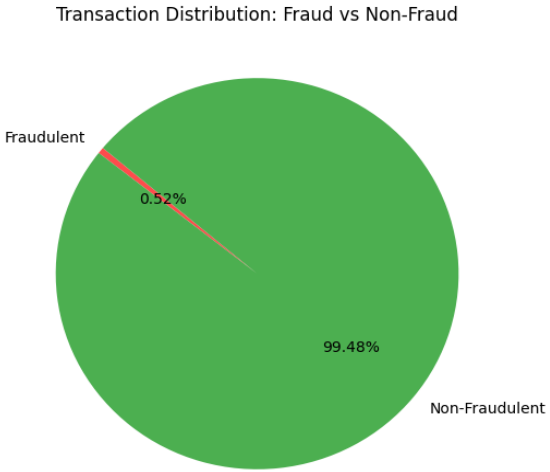
- The Problem: High manual review time and $166k/month lost to undetected fraud.

- Data Profile: 1.85M transactions, 0.52% fraud rate (highly imbalanced).

- Technique: SMOTE oversampling to

handle imbalance

and detect rare fraud patterns.

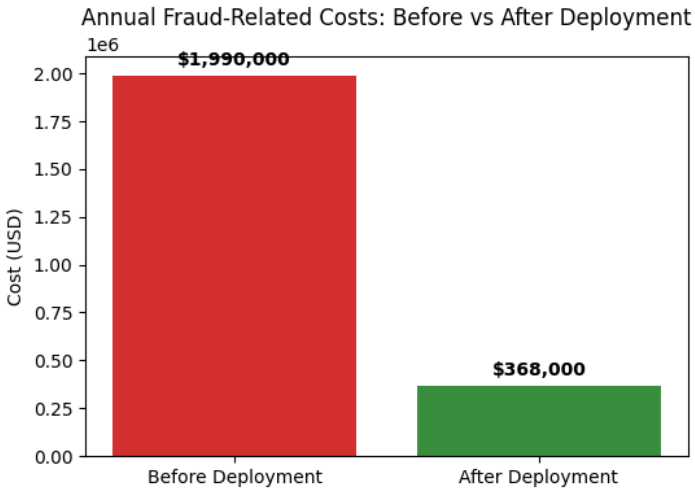


MODEL PERFORMANCE (XGBOOST ADVANTAGE)

- Model Selection: XGBoost outperformed Logistic Regression and Random Forest.
- Detection Rate (Recall): 81.77% — catches over 8 out of 10 frauds.
- Customer Impact (FPR): 0.96% — less than 1 in 100 legitimate transactions flagged.
- Visual: ROC Curve & Confusion Matrix showing AUC = 0.988.

FINANCIAL IMPACT (COST-BENEFIT ANALYSIS)

- Before Deployment: $1.99M annual loss from fraudulent claims.
- After Deployment: $368k annual cost (support + undetected fraud).
- Net Annual Savings: $1,625,957.84.



RECOMMENDATIONS & IMPLEMENTATION

- Immediate Action: Deploy XGBoost model with SMS-based 2FA.
- Timeline: 4–6 week full implementation roadmap.
- Monitoring: Weekly tracking of 0.96% false positive rate.
- Outcome: Protect bank reputation and achieve ROI within 2 weeks.
- Visual: Implementation timeline graphic.